Yahtzee Game Report

Final Assignment Y1S1 CS/CSE

27/01/2025

IT24103215

M.S.L.Narangoda

Overview

The implemented Yahtzee game is a simple version of the classical dice game. Here a human player play against the computer player which tries to beat the human player using the game strategies. In this game, a player can re-roll the dices up to three times to create the best scoring dice pattern in each round. After the final round, program automatically calculates the final scores of the both players and announces winner of the game with the highest score.

Key Features

1) Dice Re-roll Function:

- Each player can hold the dices they want and reroll the rest up to three times in each round. This allows the players to create the best scoring dice pattern.

2) Computer AI:

- Human player plays against the computer AI, which always tries to make the optimal decisions and win.

3) Score Calculation:

- Scores are calculated automatically based on the final dice configurations and selected categories.

4) Simple User-Friendly Interface:

- The game provides a clear and easy-to-follow interface for the players.

Computer Player Strategy

- **Initial Dice Roll**

After the initial roll of the dices, the computer evaluates the best possible score based on the current dice configuration. The highest score category is selected from the predefined game score categories (Yahtzee, Full House, Large Straight, etc.).

- **Selective Dice Re-Roll**

The computer calculates the frequency of each number (e.g. 1, 2, 3,...,6) and select the dices with the highest frequencies and re-roll the rest of the dices until the dice configuration matches a higher scoring category. From the predefined score categories, the lower section has the highest priority than the Upper section. So the computer try to match the dice pattern with the lower section and the upper section categories with thesholds.

- **Score Calculations**

After the final re-roll, computer calculates the scores for all available categories and choose the highest score category and assign the score to the player score array. If there are no categories to select, computer assign 0 as the score. After selecting a score category, The category is marked as used so the same category wouldn’t select again.

Challenges Faced During Implementation

- **Generating Random Numbers**

Challenge: Generating random numbers between 0 and 7 to assign as dice array values.
Solution: Used the “time.h” library to get the current time from the beginning of the UNIX epoch until now to seed the random number generator as seeding finite constant number would generate the same sequence of numbers again and again.

- **Dice Re-roll and Choosing a category**

Challenge: Determine the number of times each dice value occurs to choose a category.
Example:- Four of a kind Frequency = 4
Solution: First rearranged the dice pattern in ascending order and then calculate each number’s frequency and assign the frequency to the corresponding array position.
Example:- If the pattern contains five 4s, the frequency array would be {0, 0, 0, 5, 0}.

- **Efficiency and Accuracy**

Challenge: Ensuring the computer player makes efficient decisions without excessive computational time and make the program more easier to implement and debugging and avoid conflicts.
Solution: Create user defined functions for each calculation and categories and use structures to avoid deep nesting.

- **Scoring Category Conflict**

Challenge: Manage conflicts when multiple scoring categories are viable.
Solution: Make the Lower section has the high priority than Upper section and introduced few threshold values when computer choosing a category

- **User Interface**

Problem: Design a more user friendly, simple interface, so the human player can interact with the computer more effectively.
Solution: Human player gets clear prompts and display details about each round and expect to input numerical values to select a category(from 1 to 13), re-roll or not(1- Yes, 0- No) and selecting keeping dices.

Future Improvements

This game satisfies all the conditions that need to be done in this assignment and can further improve the following.

1. **Graphical User Interface (GUI):** Make a graphical user interface that the human player can interact with the game more efficiently and graphically.
2. **Computer AI:** Implement the computer player learns from the games and make decisions in next games other than relying on the predefined score category selecting precedence.
3. **Multiplayer Mode:** Add a multiplayer mode to the game, so more than one human player can play real time.

Conclusion

The implementation of the Yahtzee game successfully captured the core mechanics of the classic dice game and Fulfill the assignment requirements. Also, the game provides a great experience of the game and improved the programming knowledge despite the challenges faced.